



MRC Cognition
and Brain
Sciences Unit



UNIVERSITY OF
CAMBRIDGE

EEG/MEG 2:

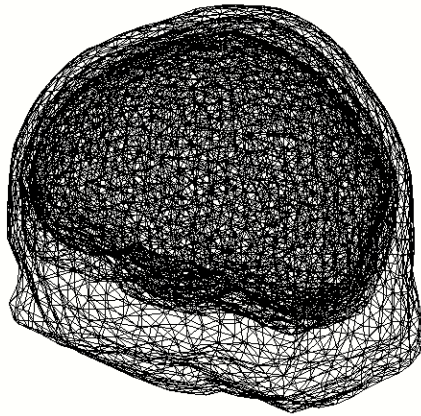
Head and Forward Modelling

Olaf Hauk

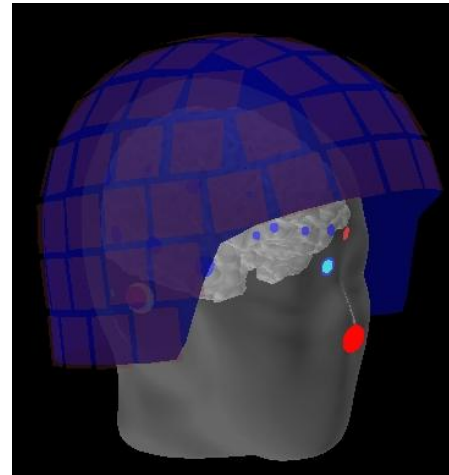
olaf.hauk@mrc-cbu.cam.ac.uk

Ingredients for Source Estimation

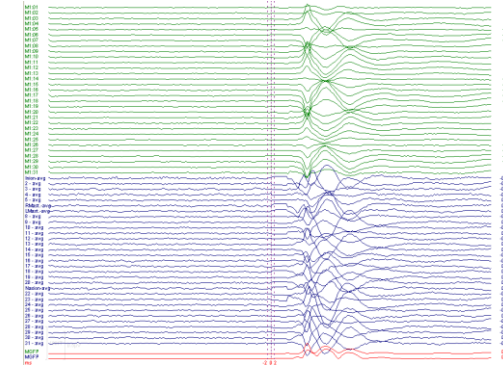
Volume Conductor/
Head Model



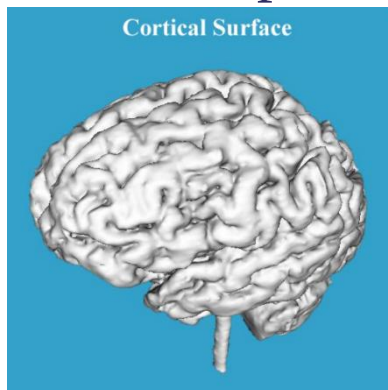
Coordinate
Transformation



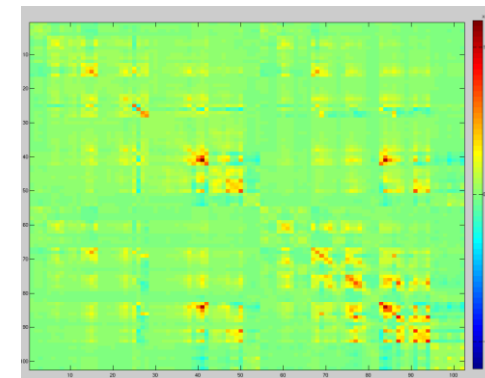
MEG data



Source Space

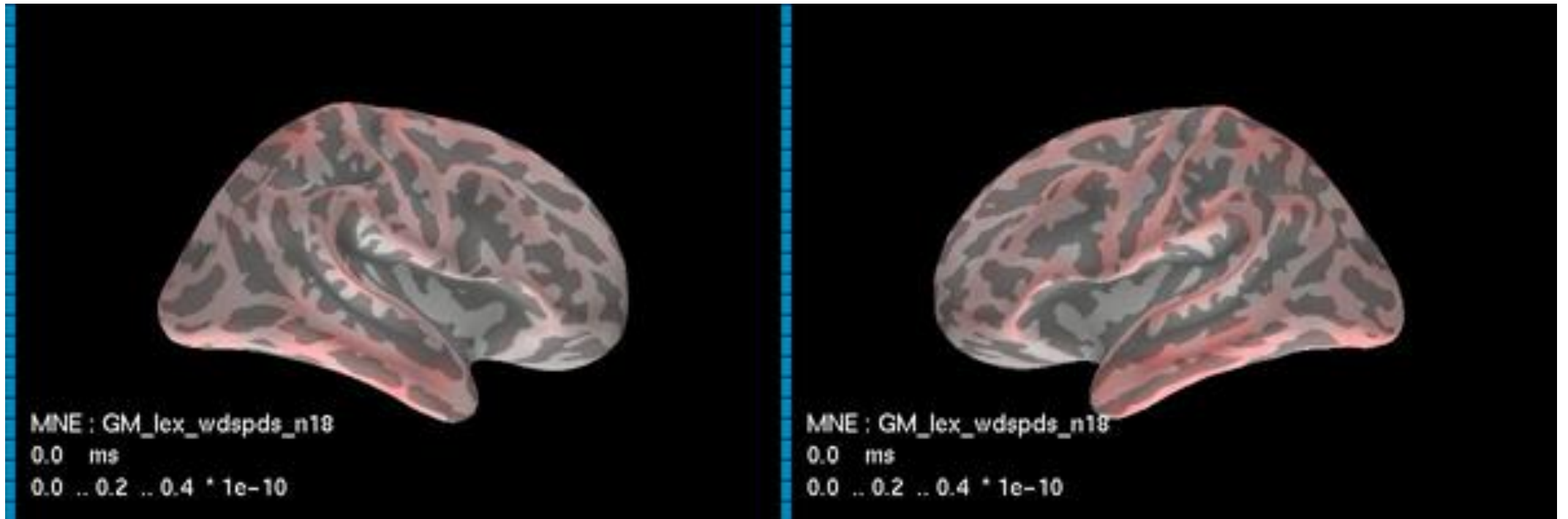


Noise/Covariance Matrix

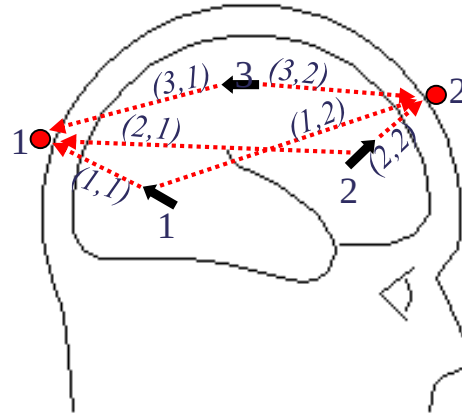


Our Goal: Spatio-Temporal Brain Dynamics

“Brain Movies”



The EEG/MEG Forward Problem



$$\begin{array}{c}
 \text{data} \quad \text{"leadfield"} \quad \text{dipoles} \\
 \begin{matrix} 1 \\ 2 \end{matrix} \cdot \begin{pmatrix} d_1 \\ d_2 \end{pmatrix} = \begin{pmatrix} 0.5 & 0 & 0.3 \\ 0 & 1 & -0.3 \end{pmatrix} \begin{pmatrix} j_1 \\ j_2 \\ j_3 \end{pmatrix} \begin{matrix} \nwarrow 1 \\ \nearrow 2 \\ \nwarrow 3 \end{matrix} \\
 \text{?} \\
 \text{inversion} \\
 \begin{matrix} \nwarrow 1 \\ \nearrow 2 \\ \nwarrow 3 \end{matrix} \begin{pmatrix} j_1 \\ j_2 \\ j_3 \end{pmatrix} = \begin{pmatrix} 1.5034 & 0.1241 \\ 0.2483 & 0.9379 \\ 0.8276 & -0.2069 \end{pmatrix} * \begin{pmatrix} d_1 \\ d_2 \end{pmatrix} \begin{matrix} 1 \\ 2 \end{matrix}
 \end{array}$$

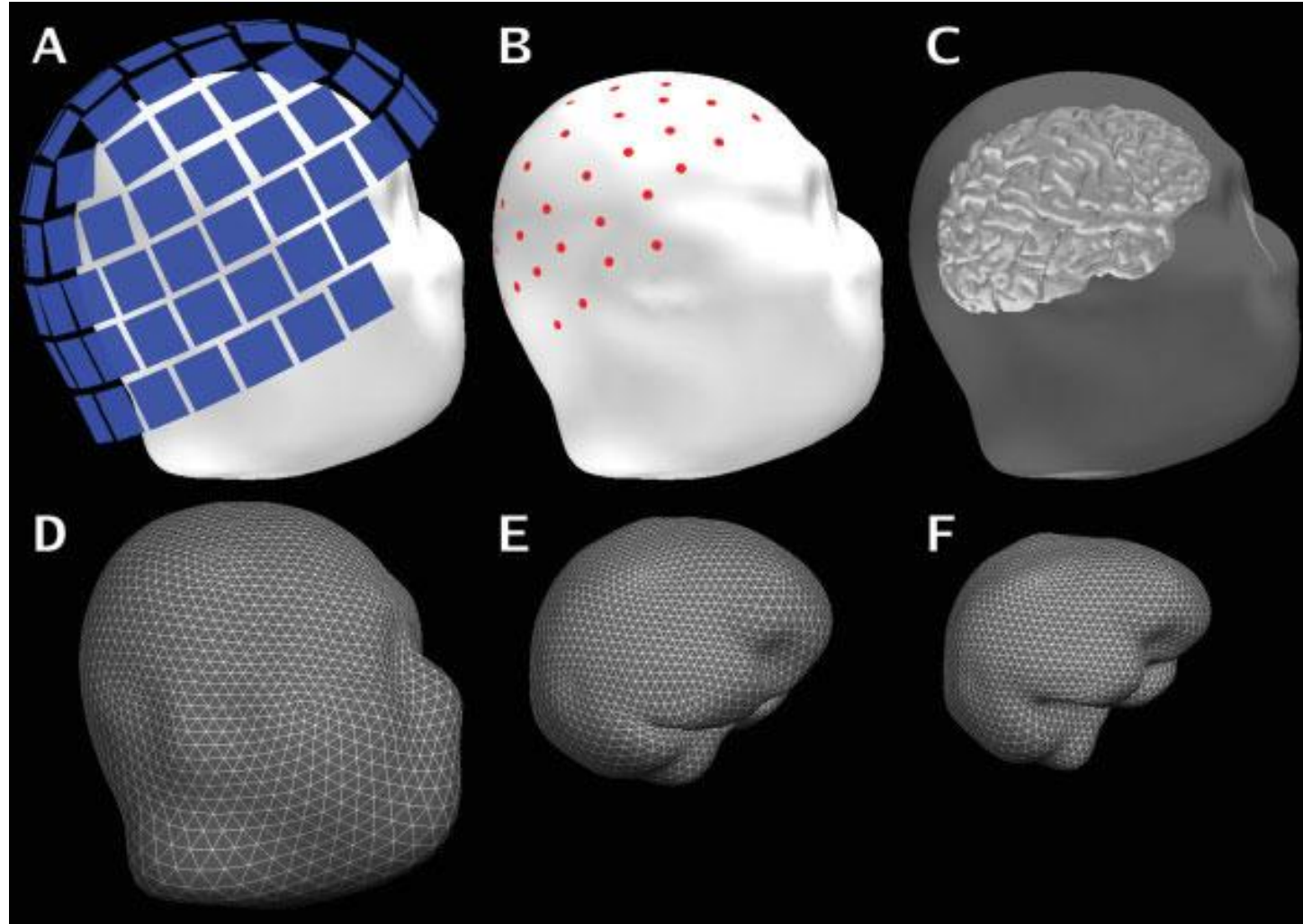
$$j_1 + j_2 = 1$$

under-determined problem, no unique solution

$$\mathbf{d} = \mathbf{L}\mathbf{j}$$

d: data (n_sensors x 1) **L**: "leadfield" (n_sensors x n_dipoles), **j**: dipoles (n_dipoles x 1)
Usually n_dipoles >> n_sensors.

Ingredients for a head model

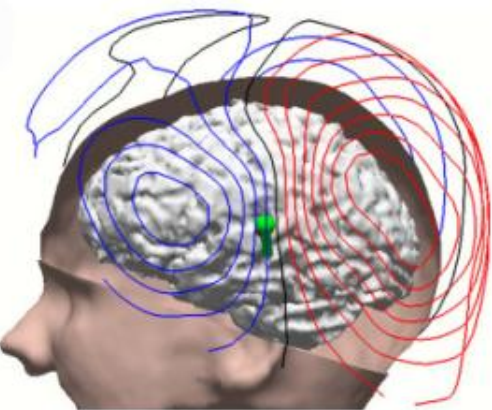


Goldenholz et al., HBM 2009
<https://pubmed.ncbi.nlm.nih.gov/18465745/>

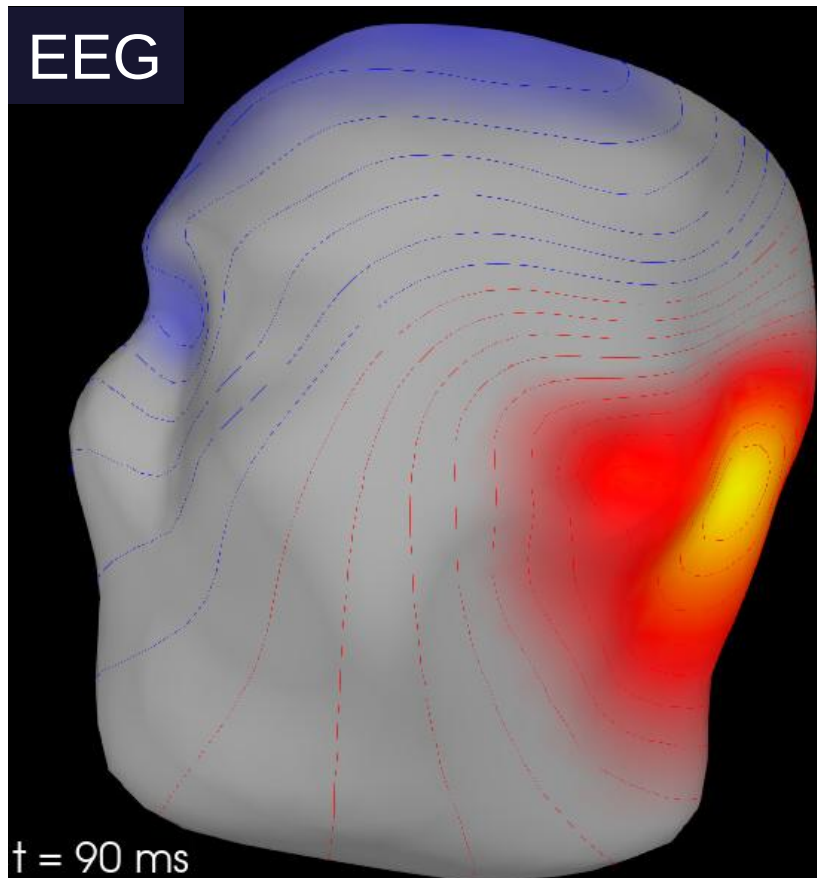
If you don't have individual MRIs: Standard head models and spherical approximations are available.

Example: Auditorily Evoked Activity

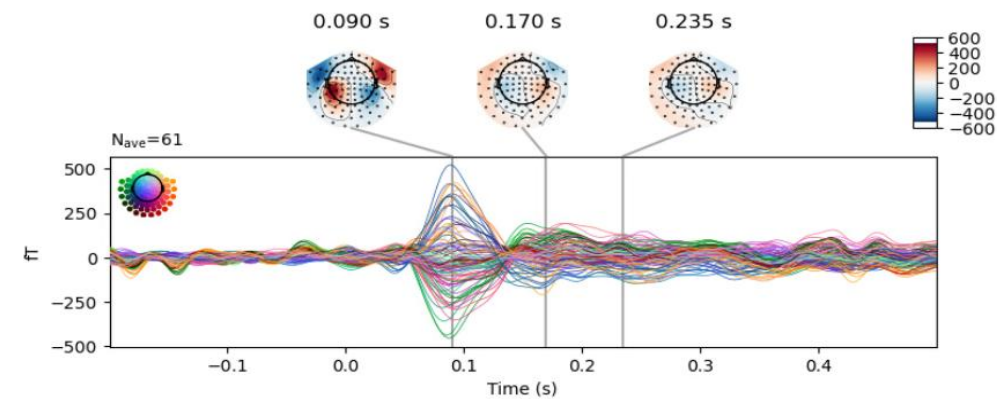
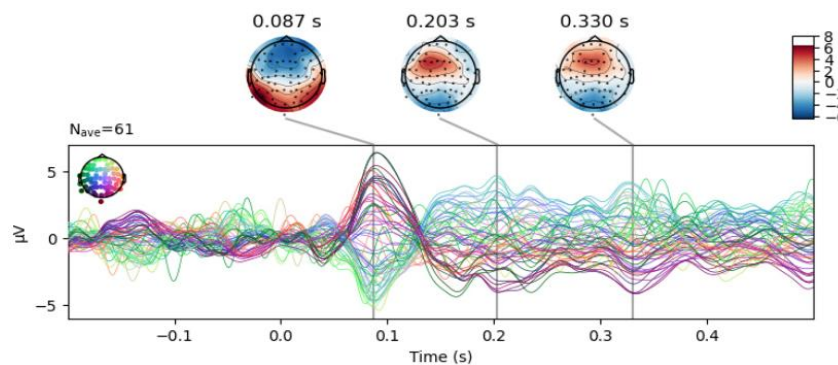
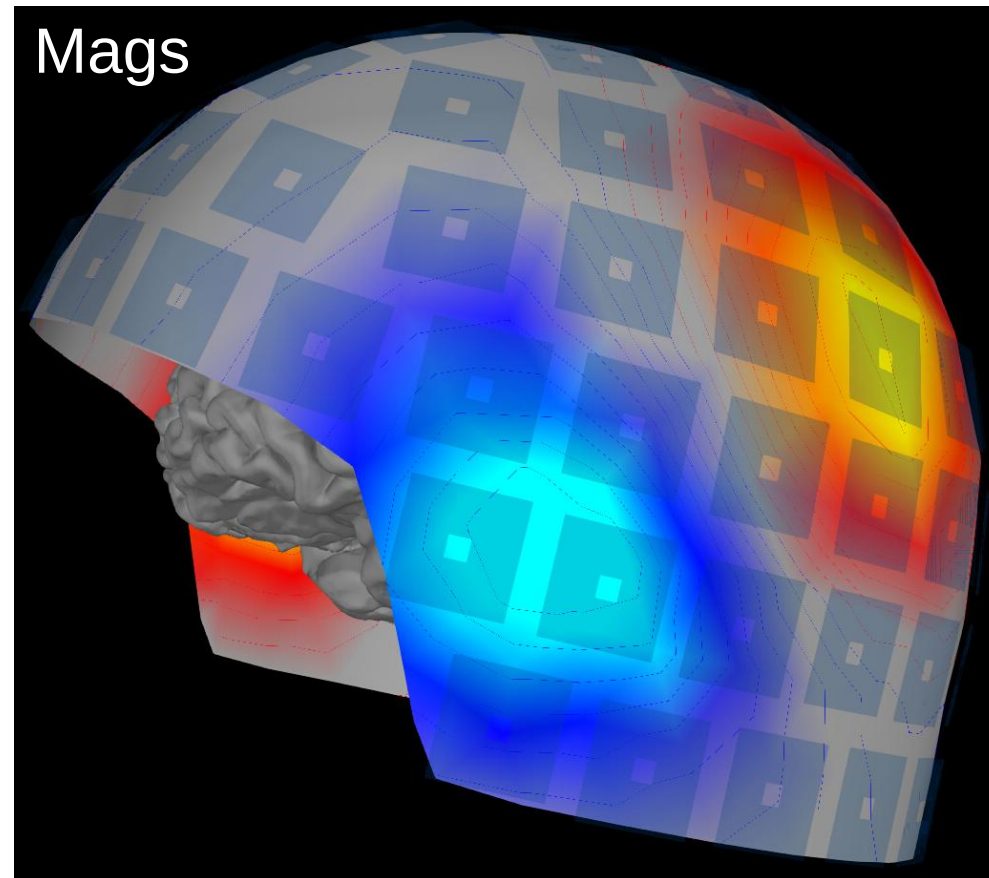
Tone to right ear



EEG

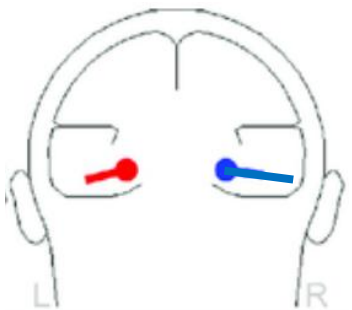


Mags

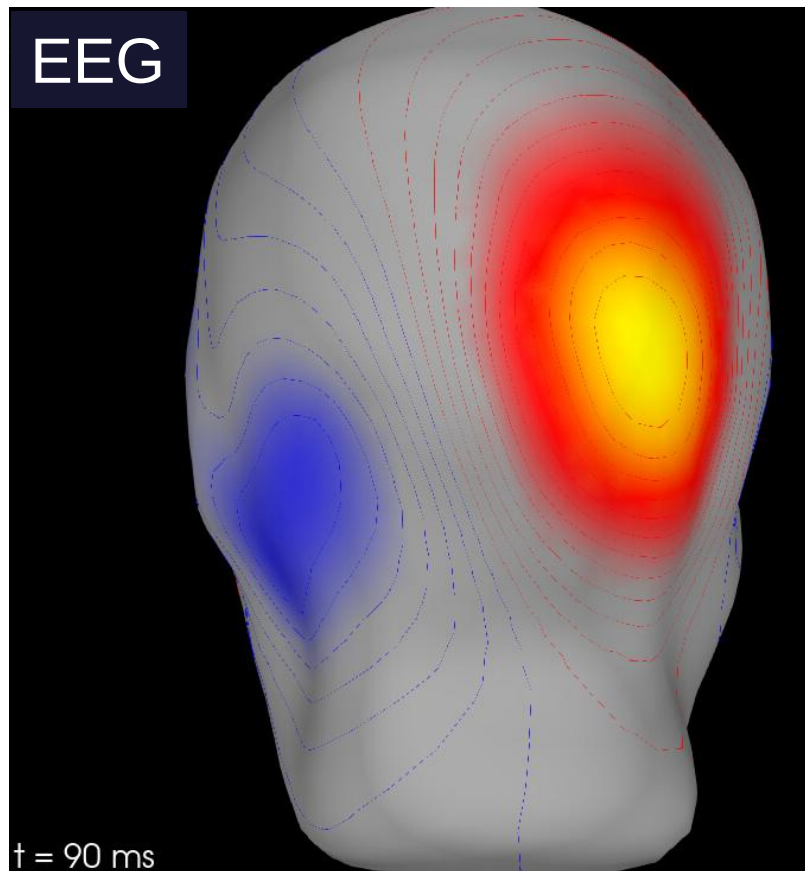


Example: Visually Evoked Activity ~100 ms

Checkerboard to
left visual field

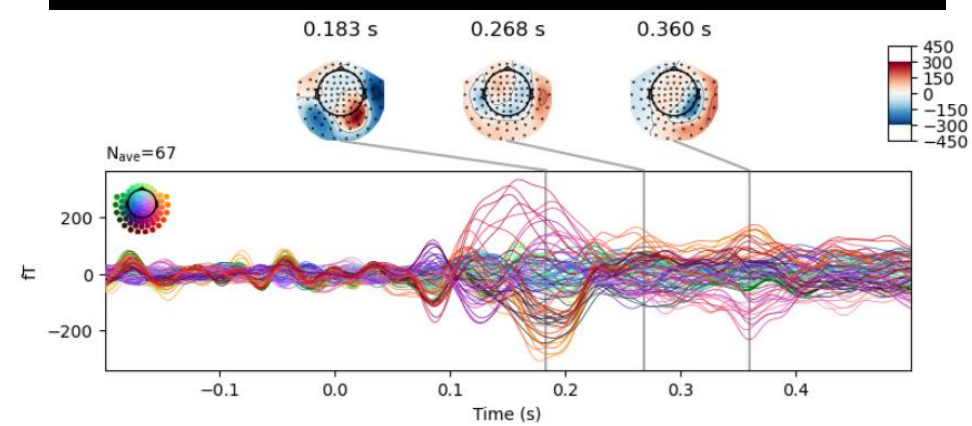
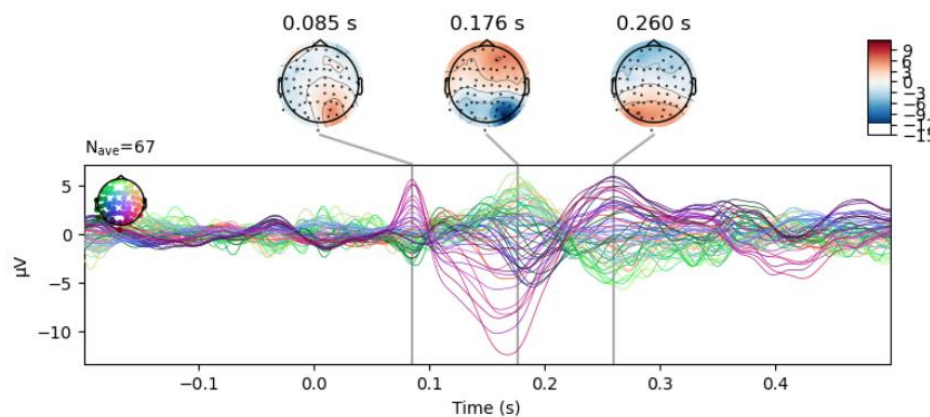
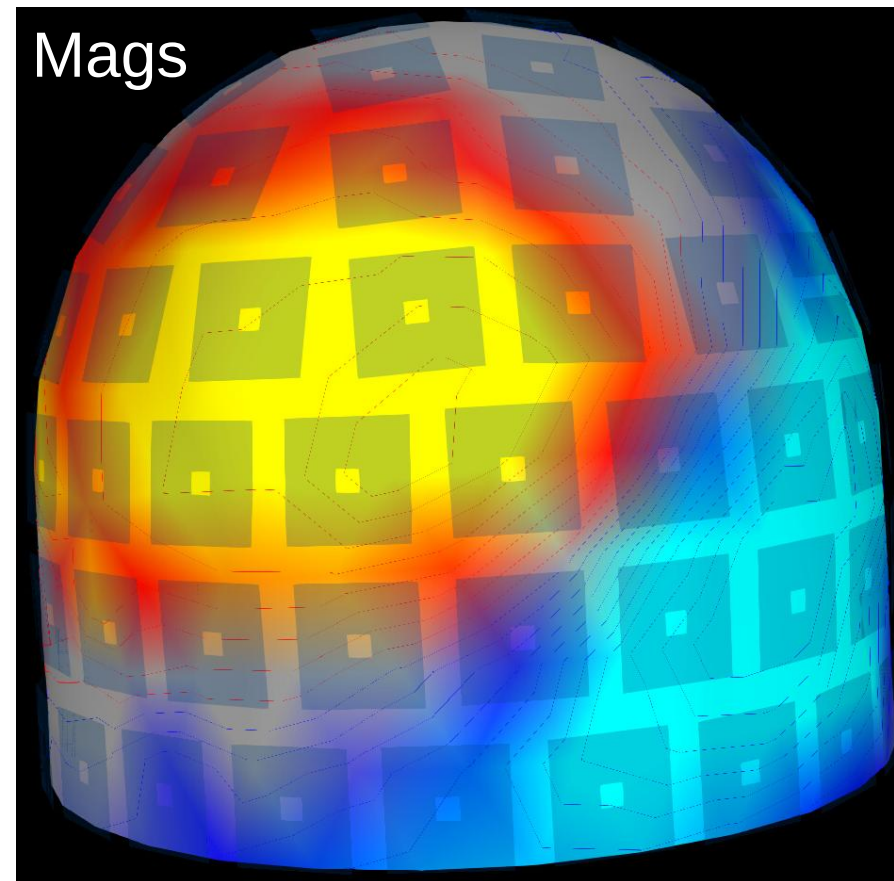


EEG



$t = 90 \text{ ms}$

Mags

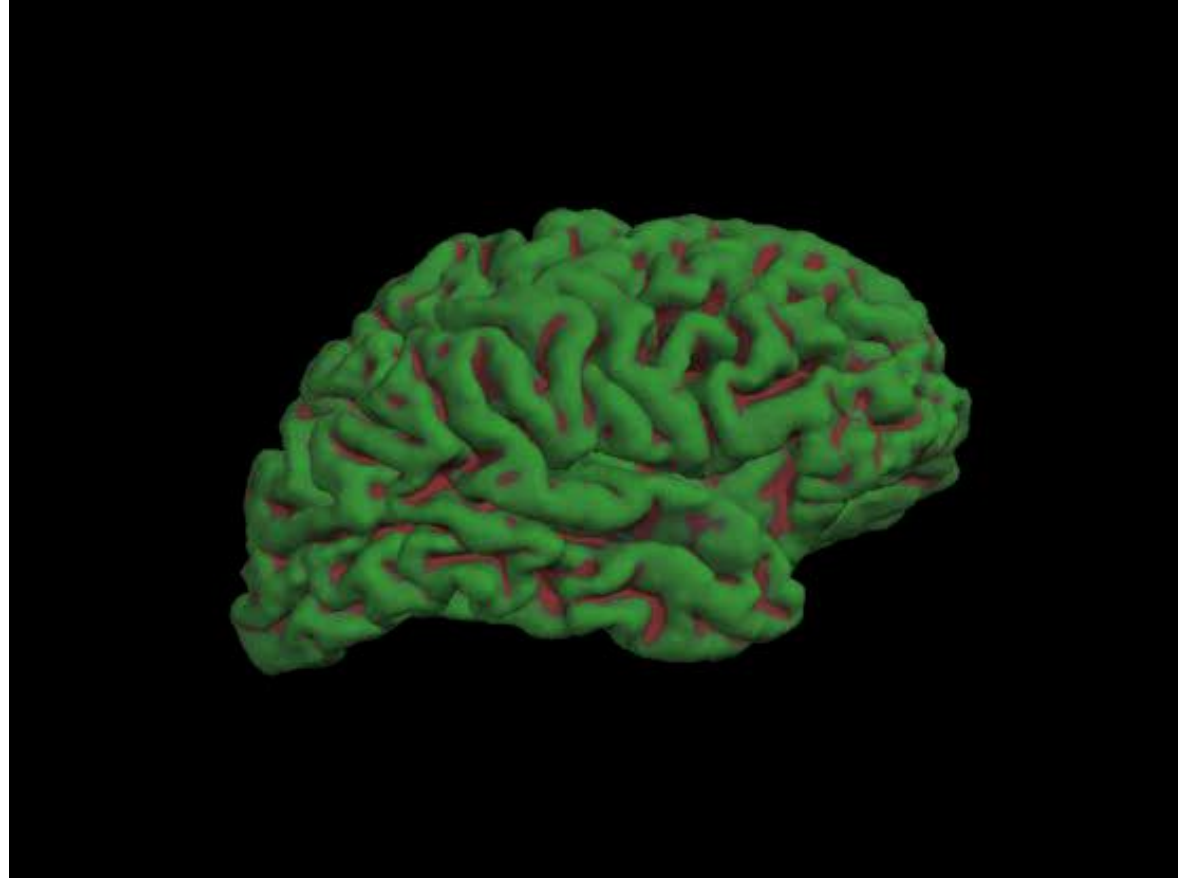


The Forward Problem and Head Modelling

Source Spaces

Source Space

Where active sources may be located, e.g. grey matter, 3D volume



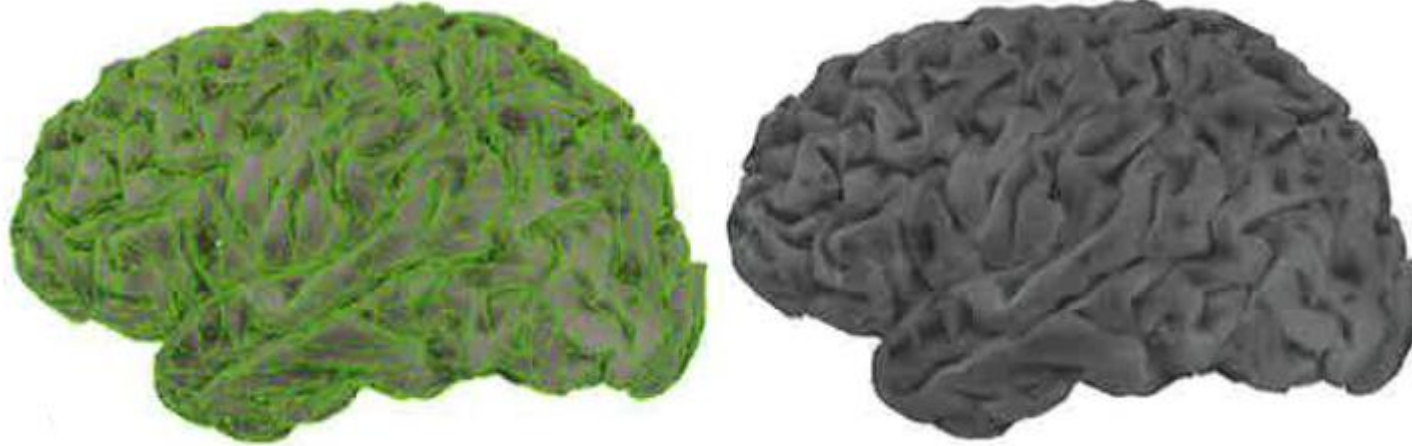
<http://www.cogsci.ucsd.edu/~sereno/movies.html>

Sometimes “standard head models” are used, when no individual MRIs available.

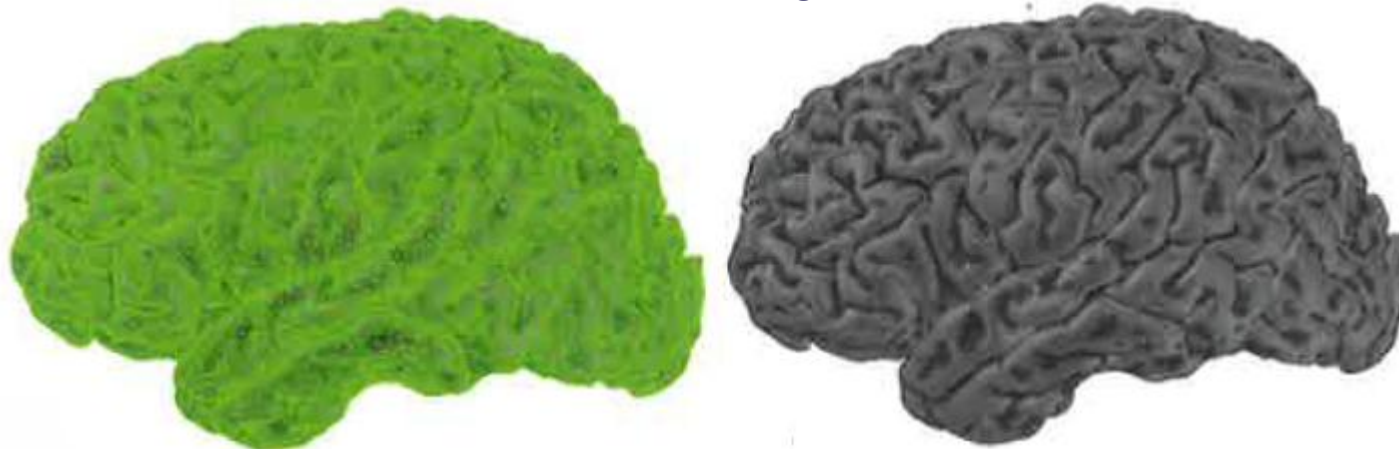
SPM uses the same “canonical mesh” as source space for every subjects, but adjusts it individually.

Spatial Sampling of Cortical Surfaces

10.034 vertices, 20.026 triangles of 10 mm² surface area
Sufficient for most EEG/MEG applications

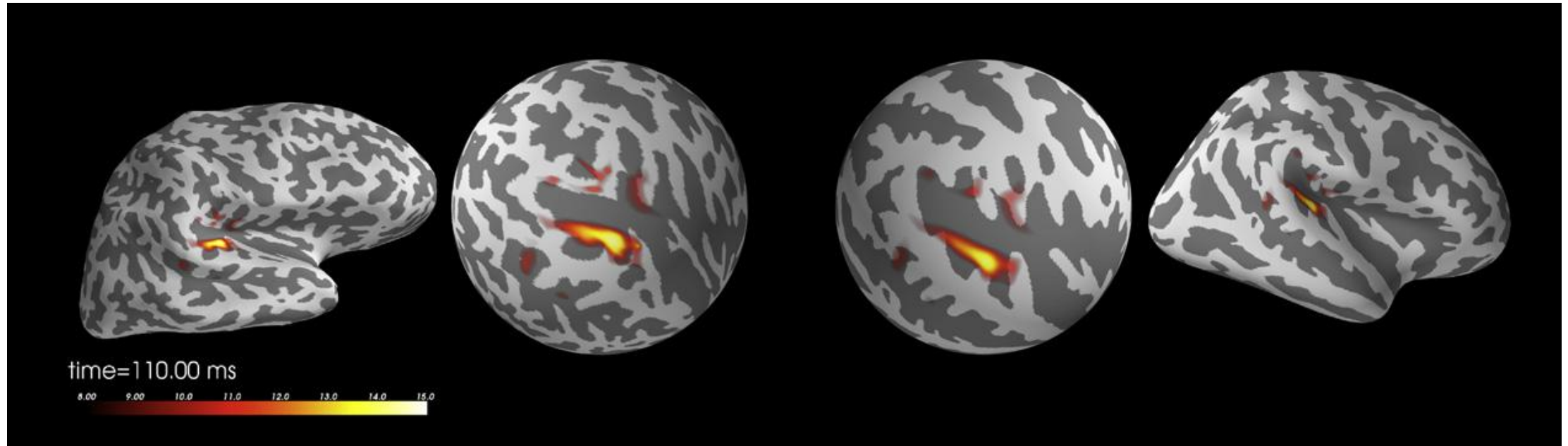


79.124 vertices, 158.456 triangles of 1.3 mm² surface area

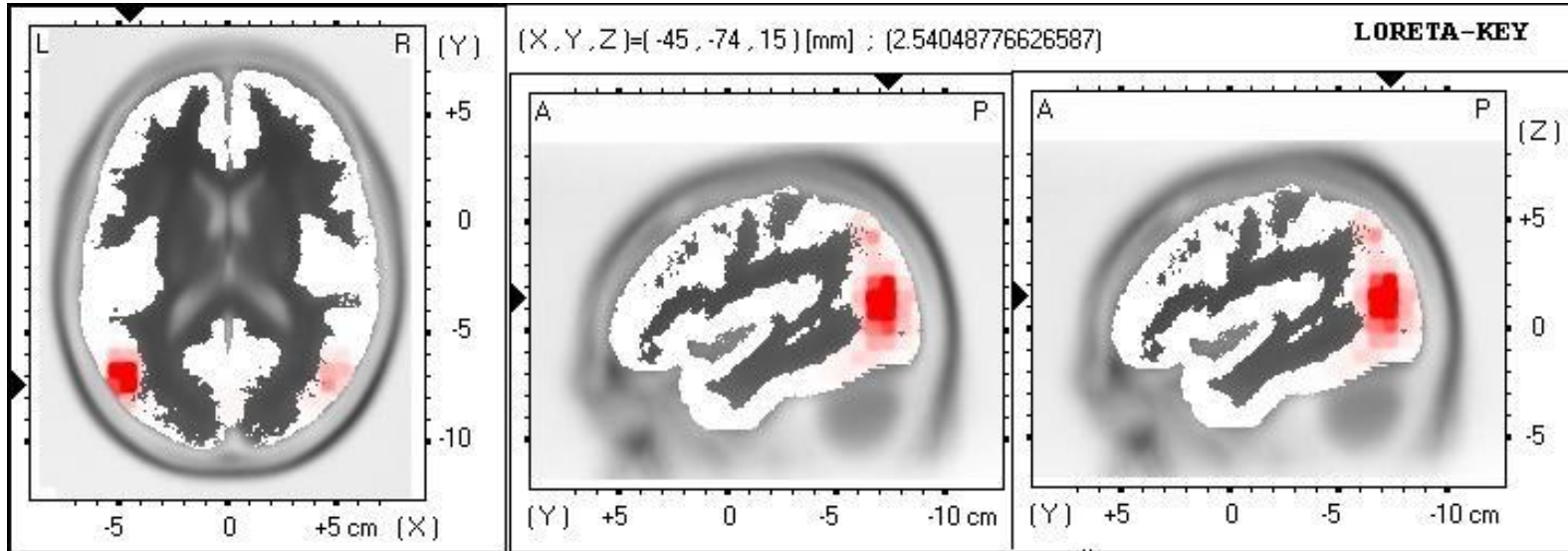


Normalising (Morphing) Cortical Surfaces

Morphing from individual to standard brain

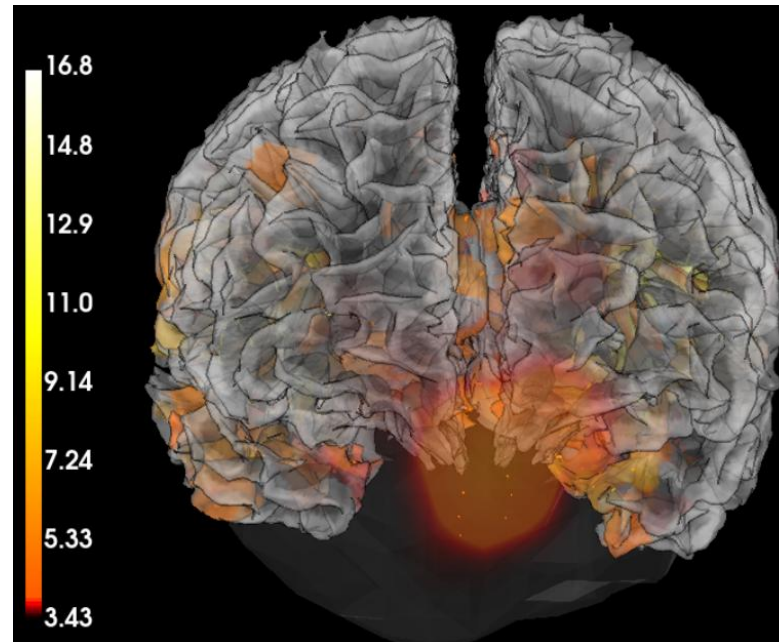
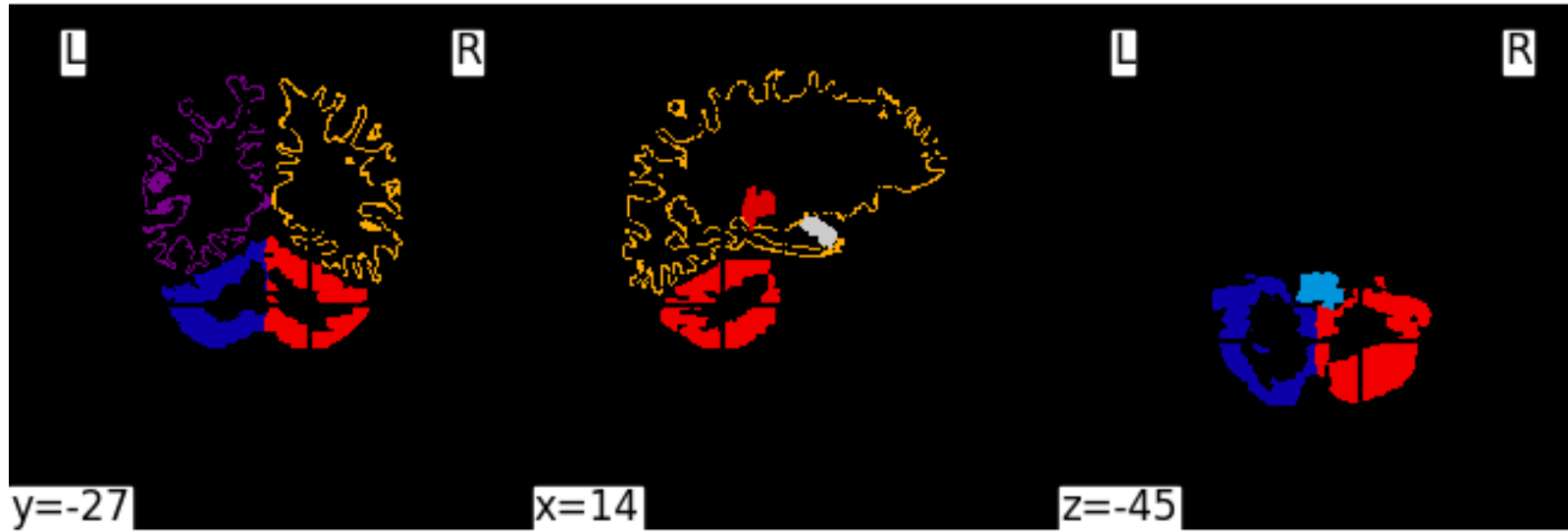


https://mne.tools/dev/auto_examples/inverse/morph_volume_stc.html



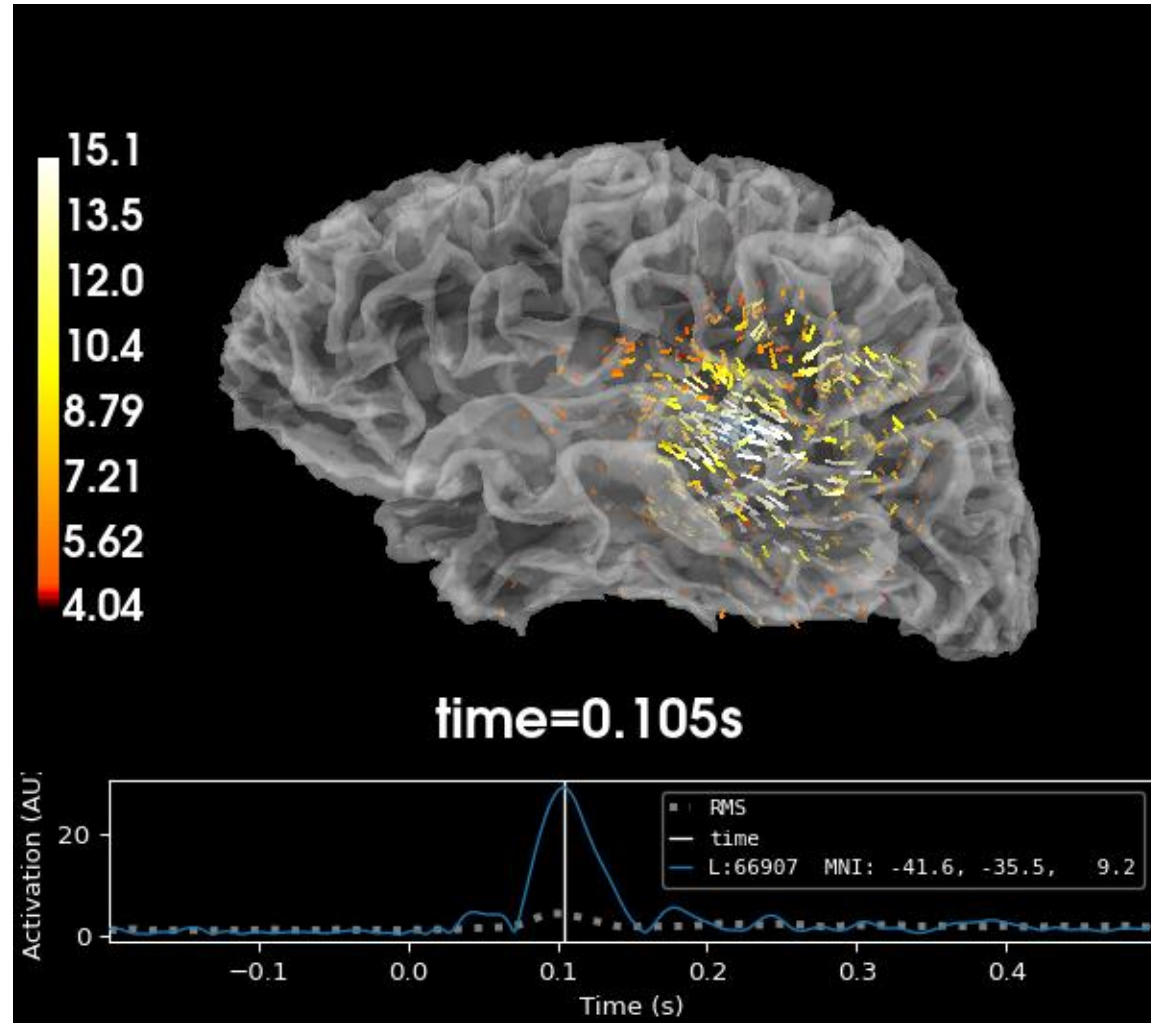
Mixed Source Spaces

Cortical and Sub-Cortical Sources

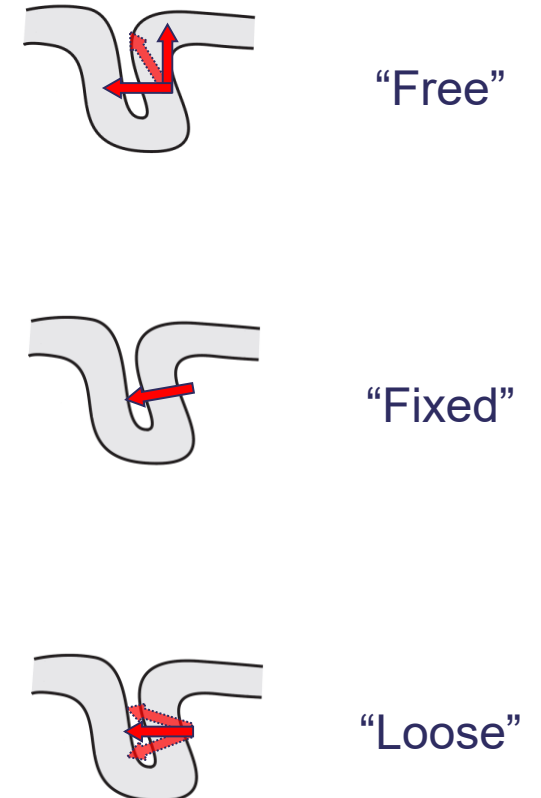


Source Orientations

Current sources have a direction and/or orientation.

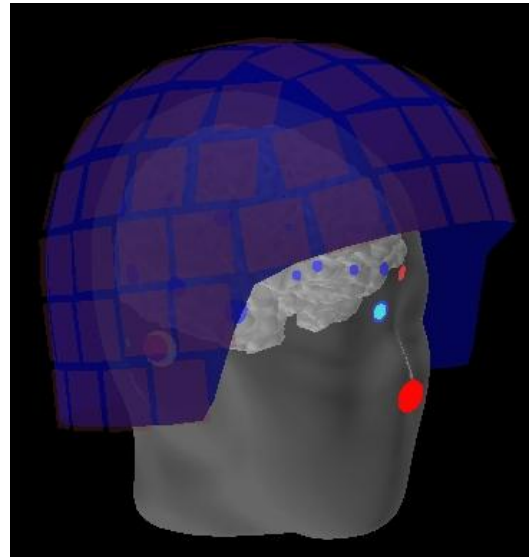


Constraints on source orientation:

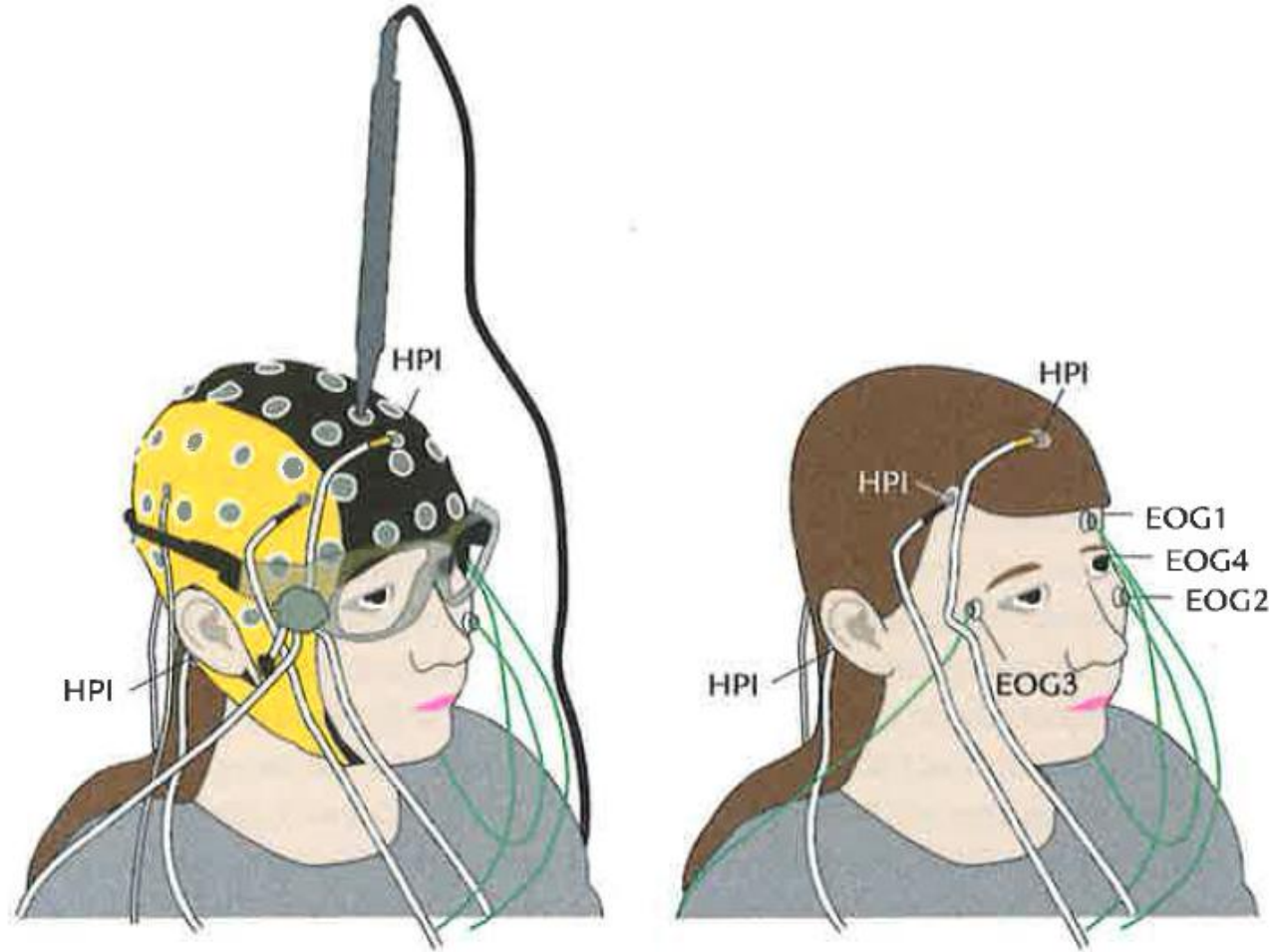


Coregistration of EEG/MEG and MRI Spaces

Coordinate
Transformation



Coregistration of EEG/MEG and MRI Spaces

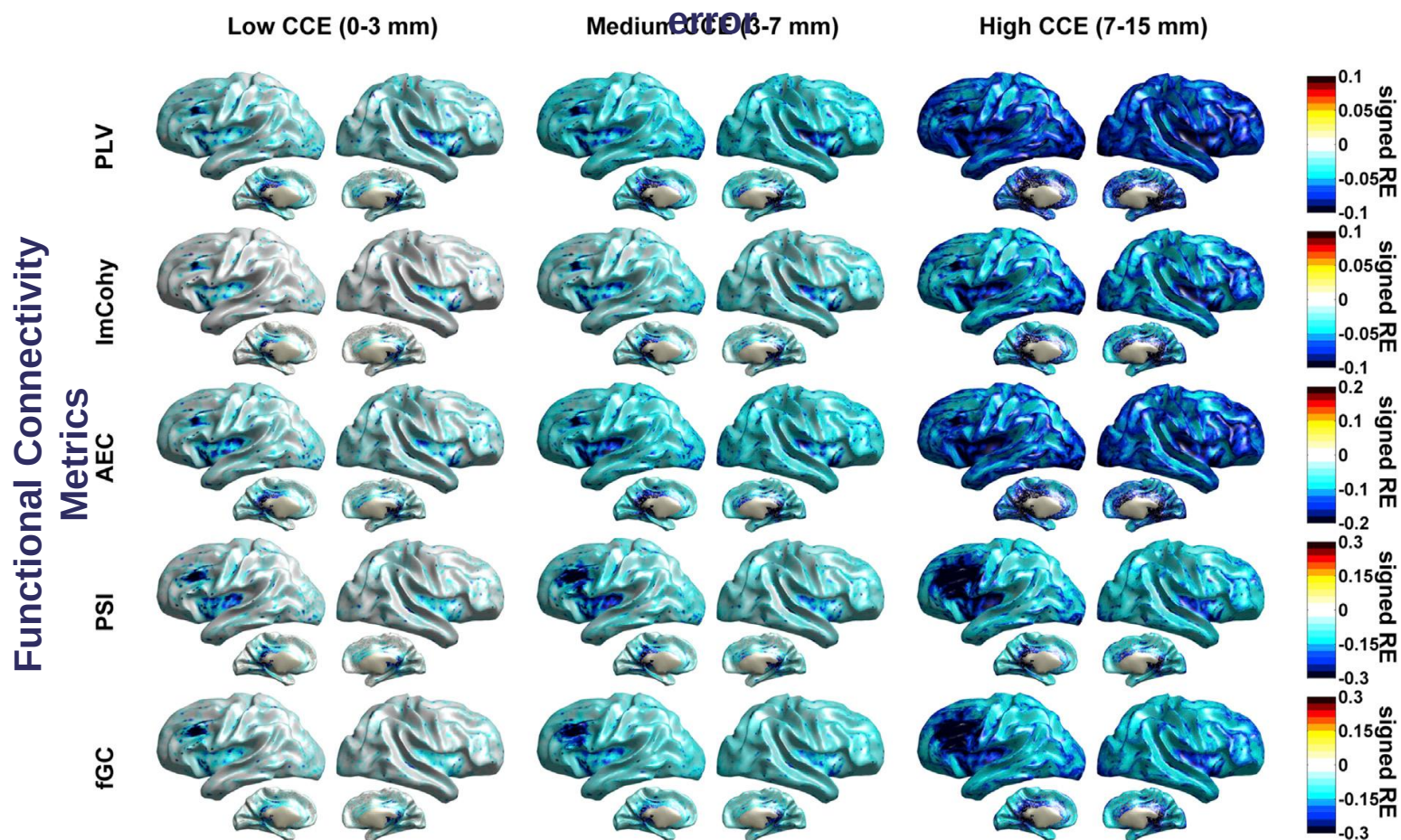


MNE-Python tutorial: <https://www.youtube.com/watch?v=ALV5qqMHLIQ>

Accurate Coregistration Is Important

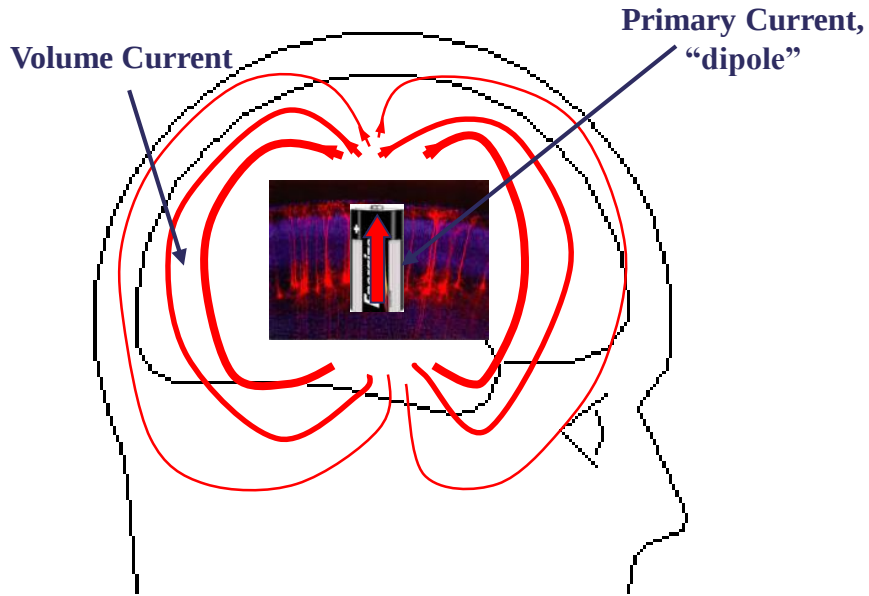
Coregistration errors affect the forward model, and therefore everything that follows.

For example, connectivity analysis:
3 levels of coregistration



Head Models

Boundary Element Model (BEM)



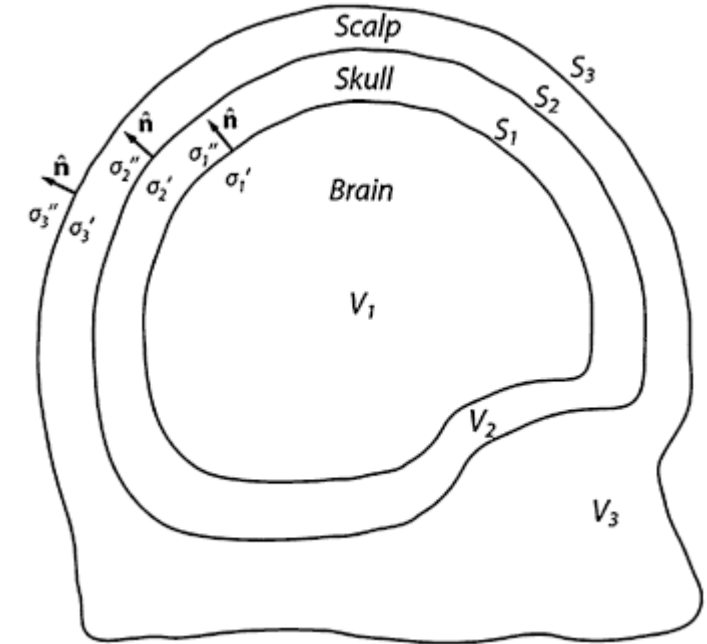
Electric potential

$$\sigma(\mathbf{r}) V(\mathbf{r}) = \frac{1}{4\pi} \sum_j (\sigma'_j - \sigma''_j) \int_{S_j} dS'_j \mathbf{n}(\mathbf{r}') \cdot \frac{\mathbf{r}' - \mathbf{r}}{|\mathbf{r}' - \mathbf{r}|^3} V(\mathbf{r}') + \frac{1}{4\pi} \int_V d^3r' \mathbf{J}^P(\mathbf{r}') \cdot \frac{\mathbf{r} - \mathbf{r}'}{|\mathbf{r} - \mathbf{r}'|^3}.$$

Magnetic Field

$$\mathbf{B}^P(\mathbf{r}) = \frac{\mu_0}{4\pi} \int_V \mathbf{J}^P(\mathbf{r}') \times \nabla' \frac{1}{|\mathbf{r} - \mathbf{r}'|} d^3r',$$

$$\mathbf{B}^R(\mathbf{r}) = -\frac{\mu_0}{4\pi} \sum_i \sigma_i \int_{V_i} \nabla' V(\mathbf{r}') \times \nabla' \frac{1}{|\mathbf{r} - \mathbf{r}'|} d^3r',$$



Heller & Volegov, in Magnetoencephalography by Supek & Aine (eds), Springer 2019

- Volume currents depend on conductivity distribution within the whole head volume.
- EEG measurements on the scalp are the result of volume currents, and are strongly affected by head geometry.
- MEG measurements are the sum of magnetic fields from primary and volume currents, but the magnetic fields of currents close to the source are much stronger than at larger distances.
- Thus, MEG signals are less affected by head geometry (e.g. skull and scalp). We usually only use one compartment (inner skull) for MEG (unless in combination with EEG).

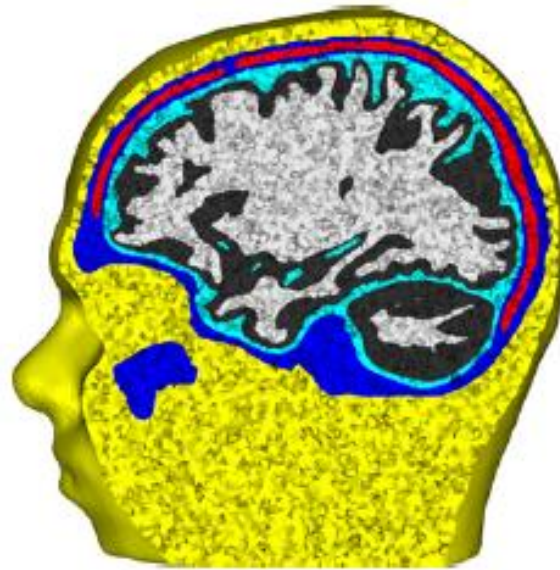
Finite Element Models (FEMs)

The use of 3-layer (brain, skull, scalp) BEM models based on individual MRI images is state-of-the-art for EEG/MEG source estimation.

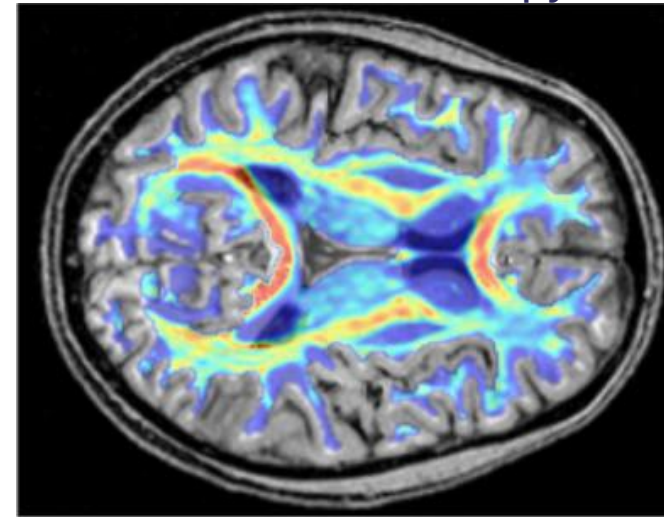
For MEG-only, single shell BEMs and local/corrected sphere models can provide reasonable approximations.

But heads are more complex:

White Matter
Gray Matter
CSF
Skull
Compacta
Skull
Spongiosa
Skin



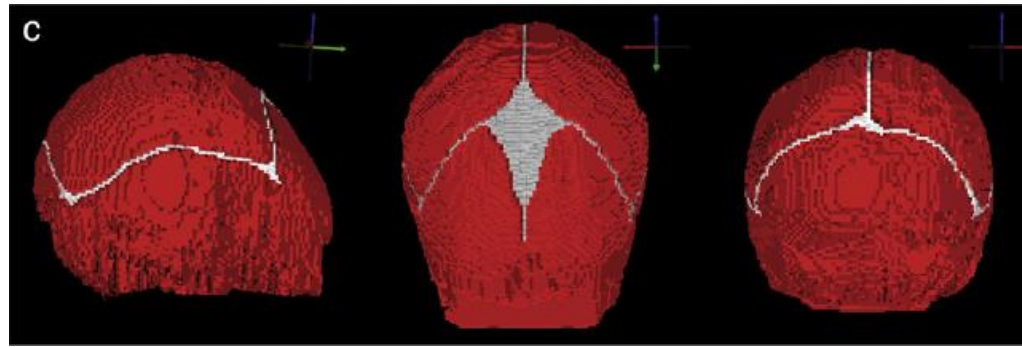
Fractional Anisotropy



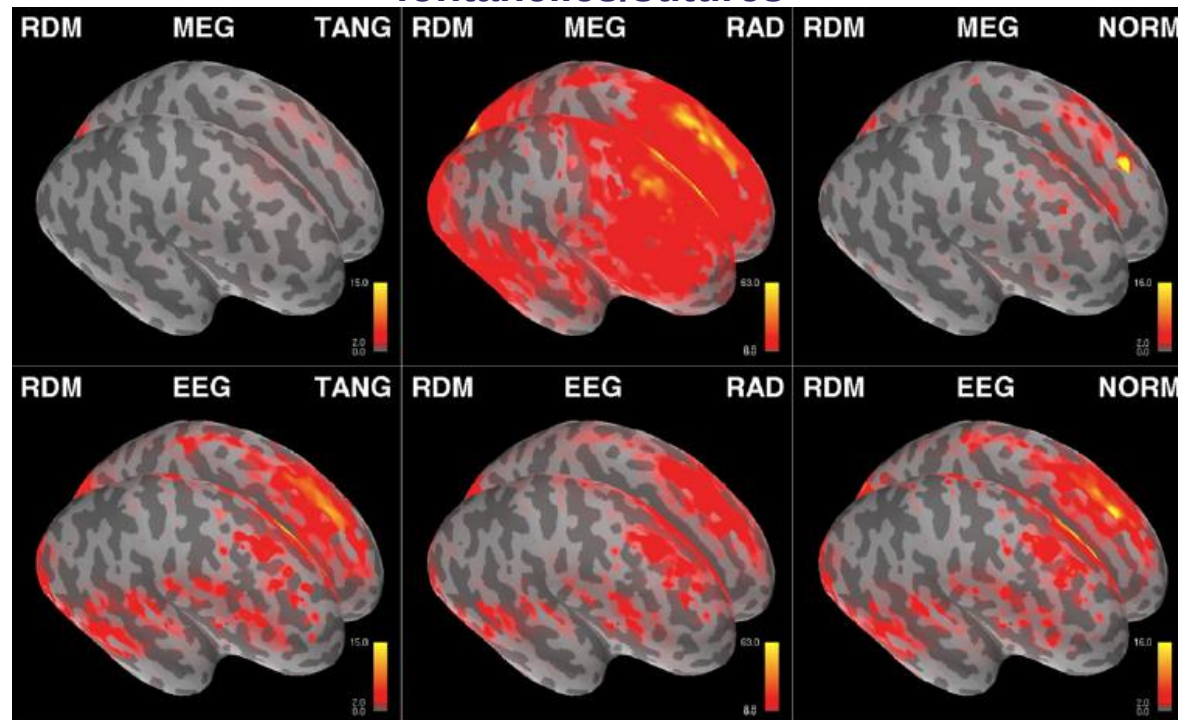
Vorwerk et al., NI 2014

It is not obvious how to translate this into more accurate estimate for conductivity distributions.

Infant Skulls – Fontanelles and Sutures



Relative error between models with and without fontanelles/sutures

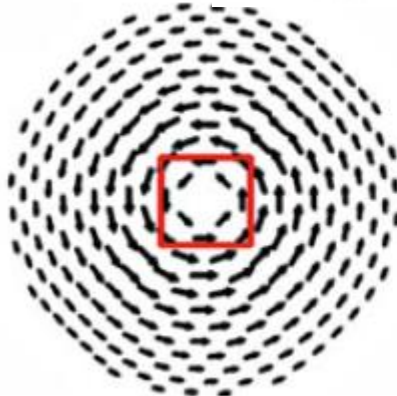
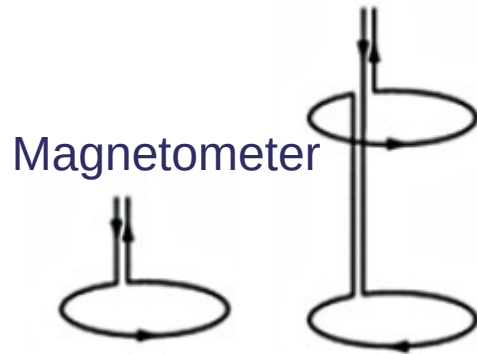


Different Sensors and their Sensitivities (Leadfields)

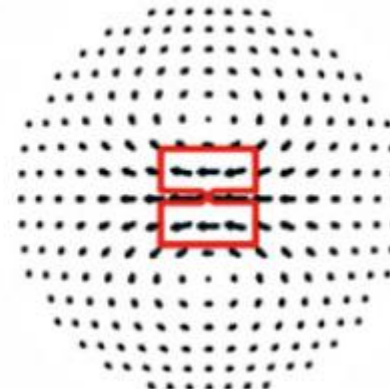
Leadfields are “sensitivity profiles” of individual sensors.

Each sensor is maximally sensitive to sources oriented along the arrows, and insensitive to sources perpendicular to the arrows.

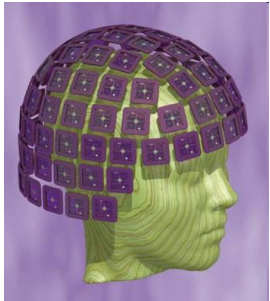
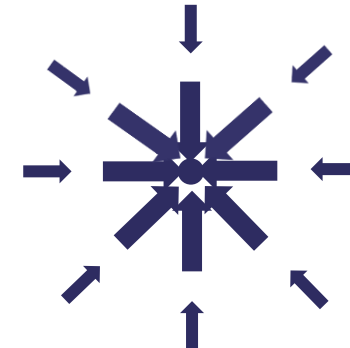
Axial Gradiometer



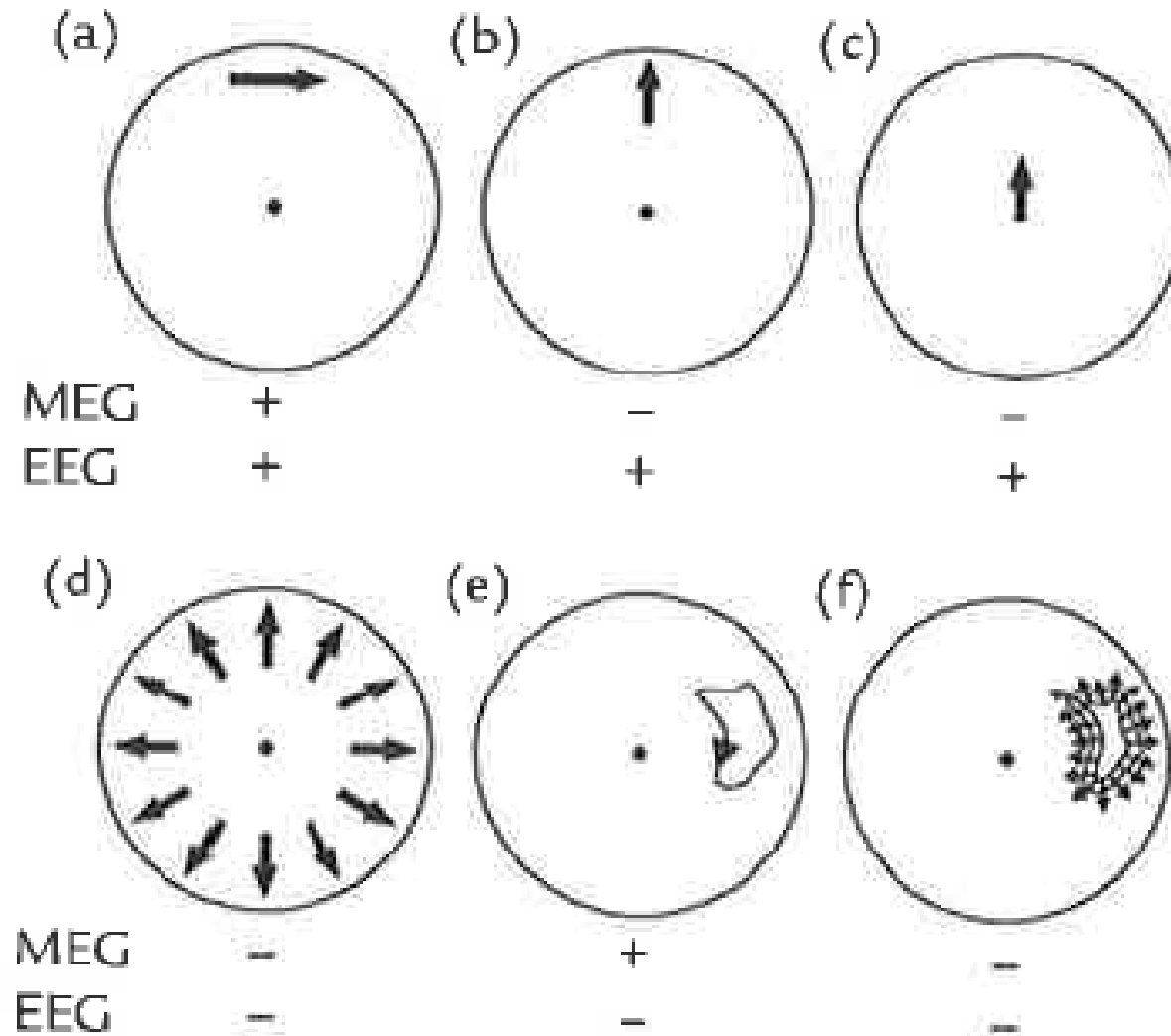
Planar Gradiometer(s)



EEG Electrode



EEG and MEG Are Differentially Sensitive To Radial and Tangential Sources

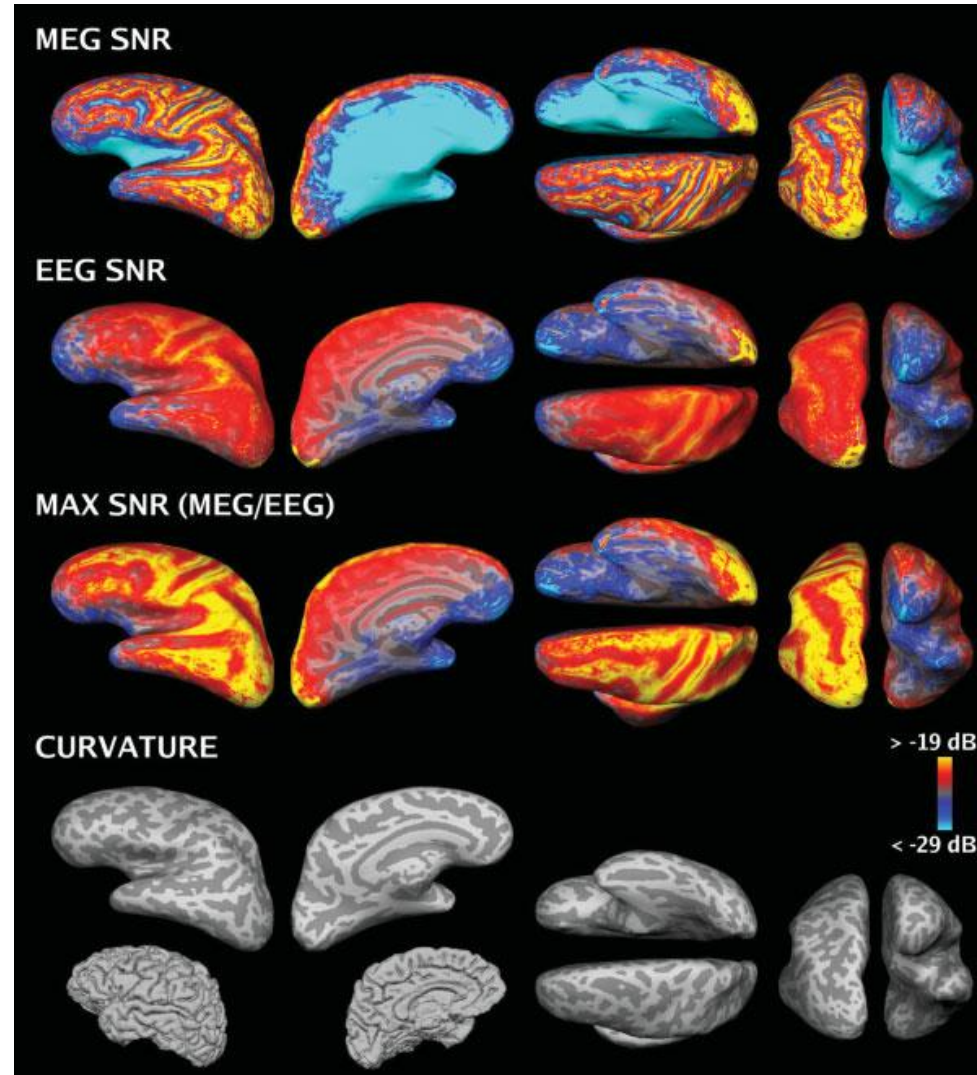


MEG is relatively insensitive to radial currents, and therefore also to deep currents.

Some complex source distributions may not produce EEG or MEG signals.

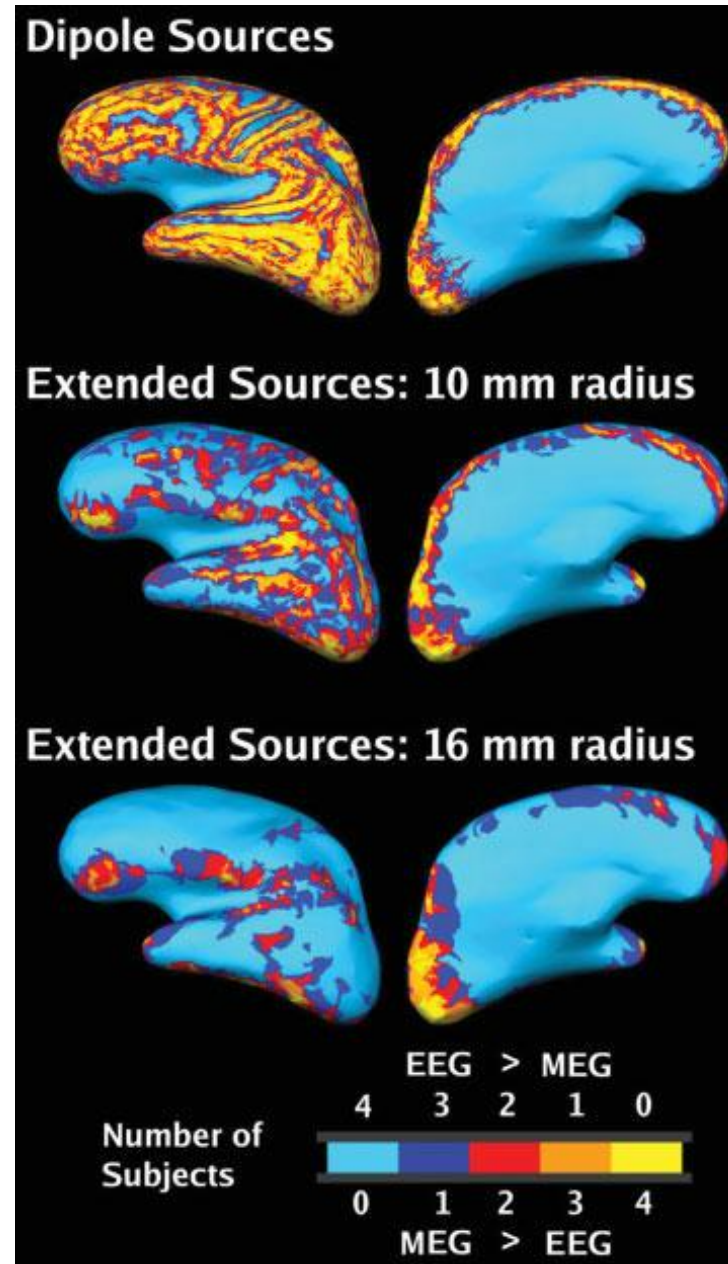
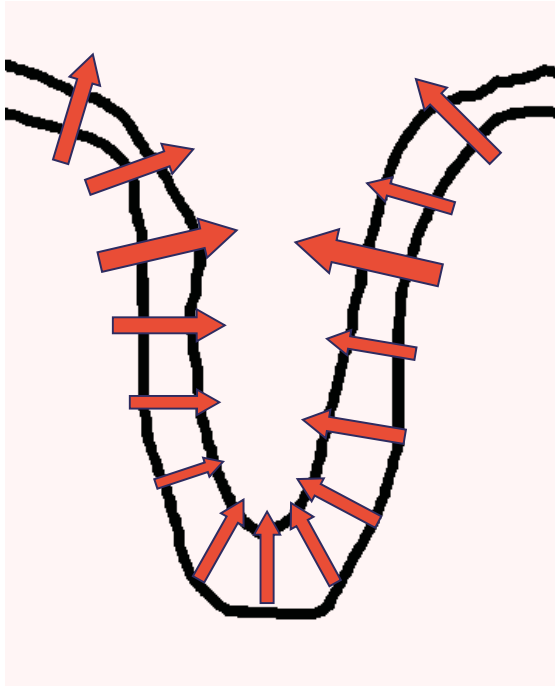
Sensitivity Maps

Sensor type, coverage and distance to sources strongly affect sensitivity and spatial resolution



MEG Is Less Sensitive To Spatially Extended Sources Than EEG

Distributed source around sulcus





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Thank you